ELECTRIC FIELDS AND ELECTRIC POTENTIALS

PURPOSE

The purpose of this experiment is:

(1) to experimentally determine equipotential lines between fixed electrodes of various geometries held at different potentials (voltages) using a digital voltmeter; and,

(2) to graphically determine average electric fields between equipotentials.

THEORY

When a potential difference (voltage difference) is applied between two fixed conducting electrodes a change in the electric potential is noted as one moves in the medium between the two electrodes.

For any given potential relative to one of the electrodes, a series of points may be found which each have the same electric potential. If these points are connected, an *equipotential line* (or surface) is determined. Since all points on this line are at the same potential, there is no electrical force to cause an electrical charge to move between any two points on such a line (that is, the electrical charge does not change its potential energy when moving along an equipotential line or surface). Note that the conducting electrodes represent equipotentials as well. On the otherhand, if a charge moves from one equipotential line to another, a force is exerted on it and its potential energy is changed.

The direction of force on a free positive charge as it moves from one equipotential line to another equipotential line (of different potential from the first) is perpendicular to an equipotential line since there is zero force and therefore zero electric field along an equipotential line. The force on a positive charge is in the same direction as the electric field line.

The direction of these E-lines is always in the direction that a free positive charge would be accelerated. *The direction is + to -, that is, from higher to lower potential.* Thus, the electric field lines are at all locations perpendicular to the equipotential lines. Once the equipotential lines are known, the E-field lines can be found graphically. While we refer to the electric field intensity at various points, since we are making somewhat coarse measurements between potential differences of 1 V or 2 V, here we are really measuring the *average electric field intensity*, $E_{avg} = -\Delta V/\Delta r$.

EXPERIMENTAL PROCEDURE

The arrangement for locating points of equipotential is shown in Figure 1 which also provides the wiring diagram for the experiment.

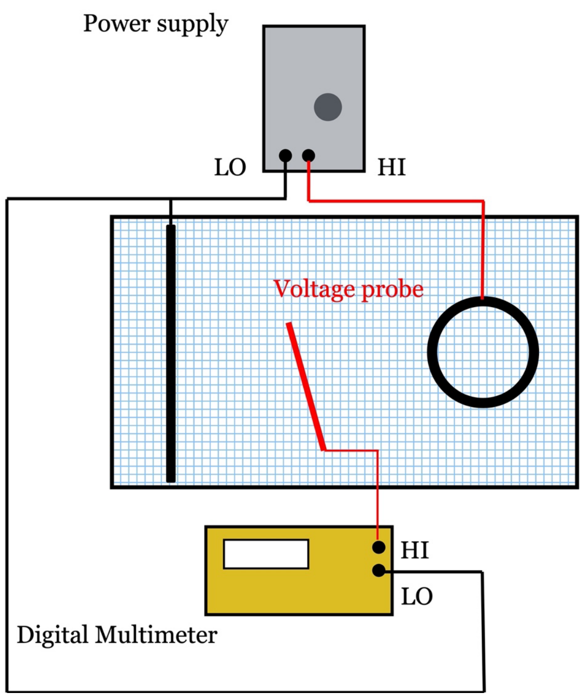

Figure 1

A transformer in the power supply takes the 120 V_{ac} line voltage and reduces it to approximately10 V_{ac}. We then adjust it to exactly 10 V with a potentiometer. This voltage is referred to as "V." (Alternating current (ac) is used instead of direct current (dc) to avoid electrode polarization effects.) We apply 10 V across the electrodes which will be of various shapes in the experiment. This voltage sets up an (alternating) current flow in the **distilled water** between the electrodes. The magnitude of the current depends on the voltage applied and the effective resistance of the water between the electrodes.

The equipotential lines are now determined in the following manner:

- (Be sure distilled water is used.)
- The multimeter "low" terminal is connected to one of the electrodes which we arbitrarily take to be a potential of 0 V.
- The other electrode is at 10 V_{ac} (consider it to be +10 V) and points in between which we measure with the probe connected to the "high" terminal of the voltmeter range between 0

and 10 V.

- The multimeter *range* should be set at 1000 V (we are safely using only up to ~15 V for the experiment), and *digits* should be set to have three decimal places displayed.

The object is to plot equipotential lines for several incremental voltages. We determine points of equal potential in the area for potential ranging from 1 V to 9 V in increments of 2 V. Once these points are determined on graph paper, we can draw lines joining points of the same potential. Several such equipotential lines are then determined.

A. TWO RING ELECTRODE CONFIGURATION

1. Typically graph paper is attached to the underside of the tray and fasten with tape.

2. Fill the tray to ~1 cm depth of distilled water.

3. Center one ring electrode at (2,9) and the other ring electrode at (22,9). Connect the two electrodes as in Figure 1 which shows wiring for ring and strip electrodes.

4. Holding the probe to the HI voltage electrode, et the potential difference between the electrodes at 10 V by adjusting the potentiometer.

Commented [E1]: Could add in "while holding the probe to the positive electrode ring"

5. Start with 1 V. Move the probe along the centerline connecting the electrodes until 1 V is read on the voltmeter. Record the (x,y) coordinate directly on another piece of graph paper on which you also have recorded the coordinate system and drawn in the electrodes.

6. Move the probe and record other coordinates which are also at a potential of 1 V (in general, space the points about 1 or 2 cm apart).

7. Repeat the above measurement procedure for 1, 3, 5, 7, and 9 V equipotentials. Also record voltages inside the ring electrodes.

8. Draw the equipotential lines for the electrodes and the five voltage values and label each line with the corresponding voltage.

9. Draw the following E-field lines indicating direction:
   a) between centers of the rings;
   b) starting off from the rings at 30° and 60° from the horizontal (note that the potentials inside the rings are constant so that the field lines are only to be drawn on the outside of the electrodes);
   c) at other appropriate angles for the lab worksheet.

B. ONE RING - ONE STRIP ELECTRODE CONFIGURATION

1. Center the ring electrode at (2,9) and place the strip on the line x = 22 with its center at (22,9). The strip electrode will be connected directly to the transformer (0 V) and the ring will be connected to the slider of the potentiometer. Check setting of 10 V between electrodes.

2. Repeat the procedure from Part A (see Figure 1).

WORKSHEET

Complete the accompanying Laboratory Worksheet